

The Economics of Science in the US

What are the relationships between these entities?

(Funding, Advises, Reporting, Programming, Appoints/Directs, Policy, Lobbies)

The Profession

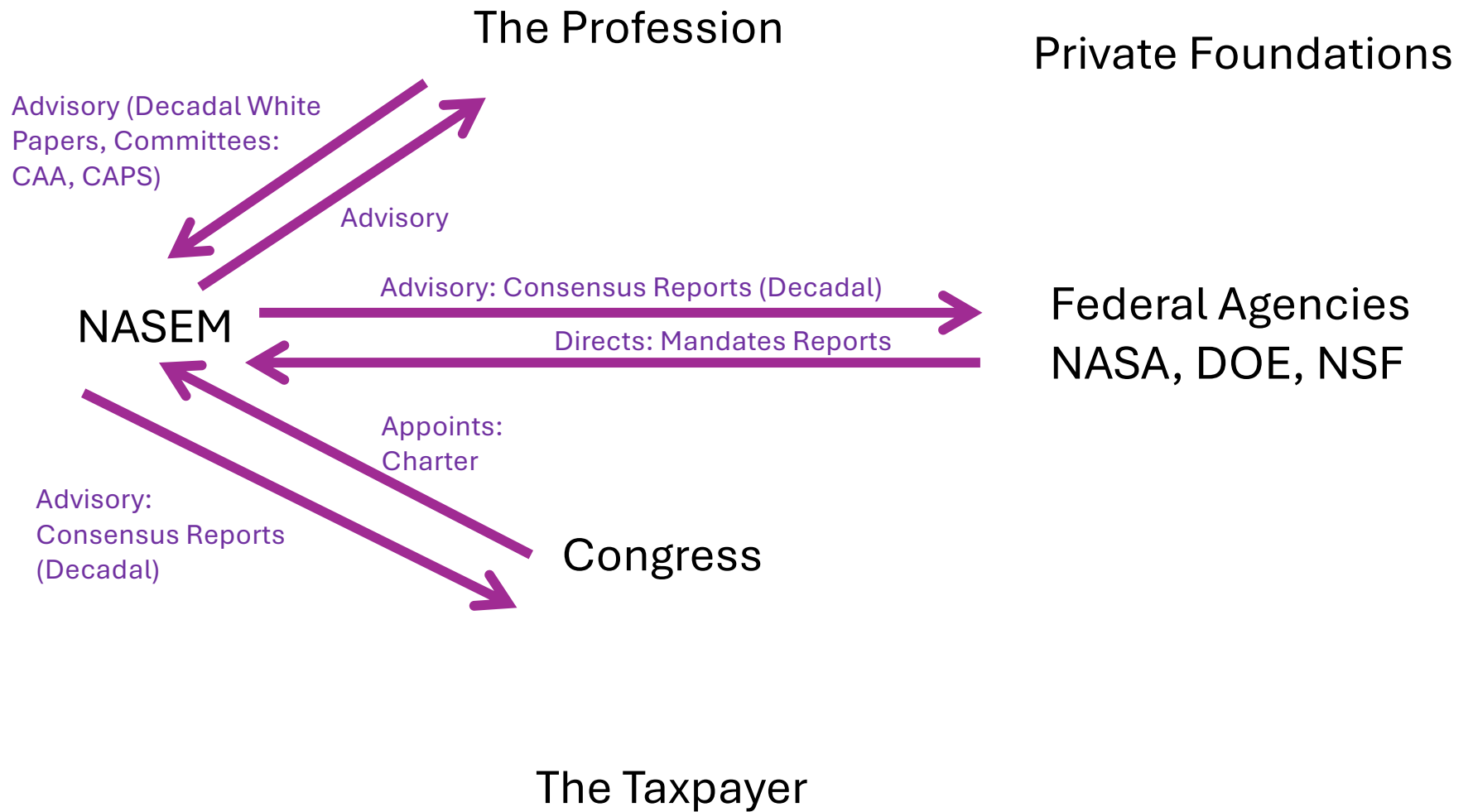
Private Foundations

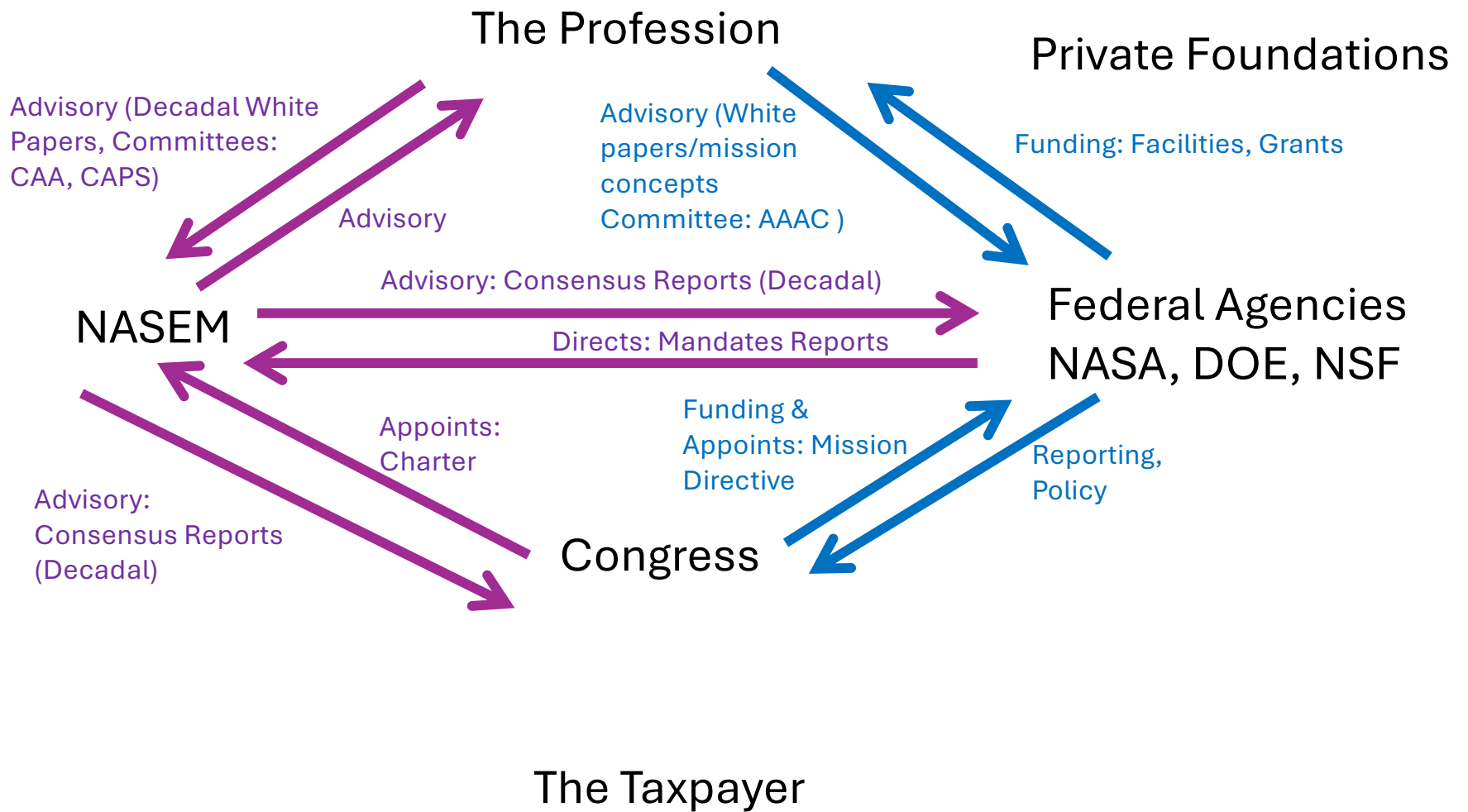
National
Academy of
Science,
Engineering
and Medicine
(NASEM)

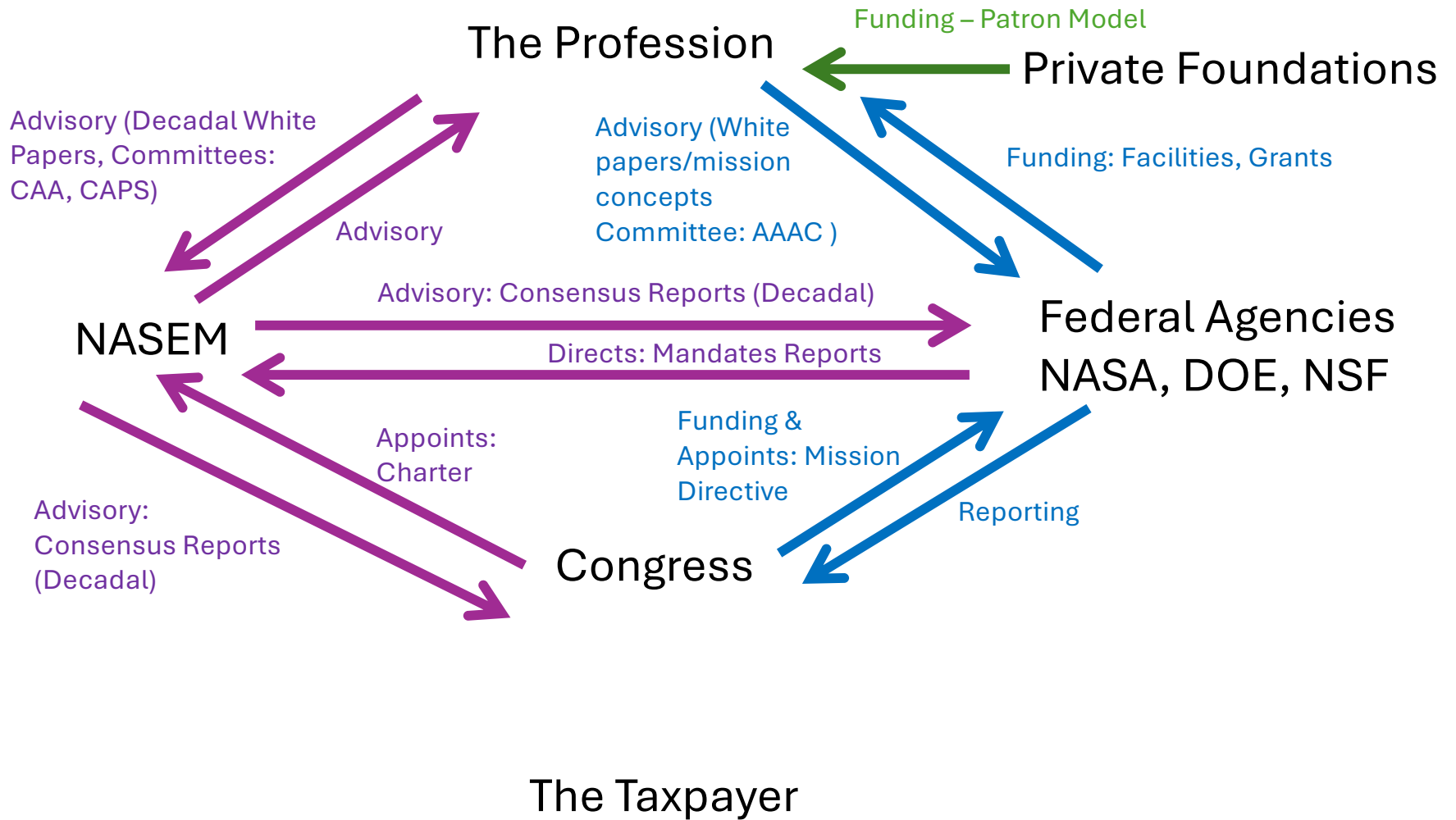
Congress

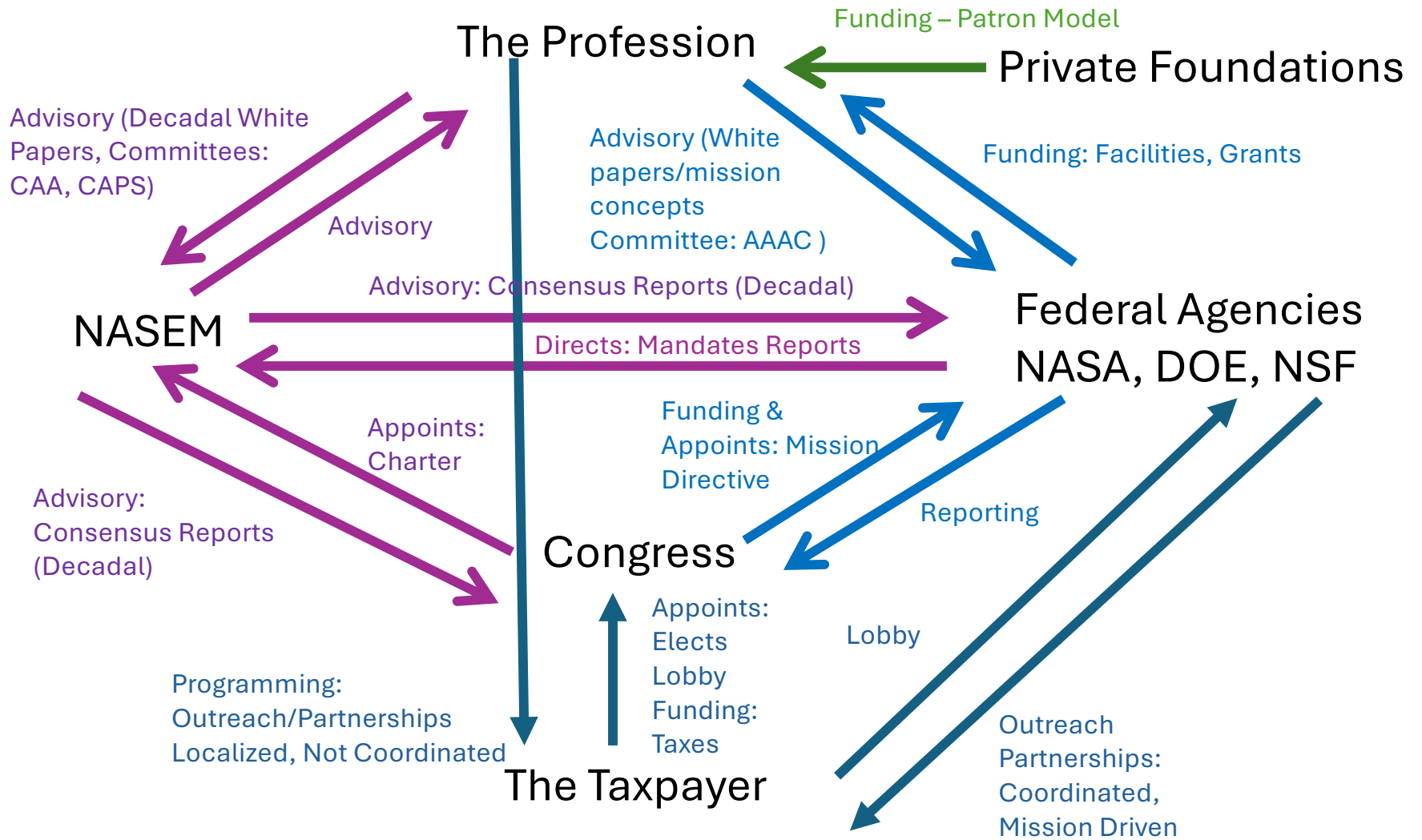
Federal Agencies
NASA, DOE, NSF

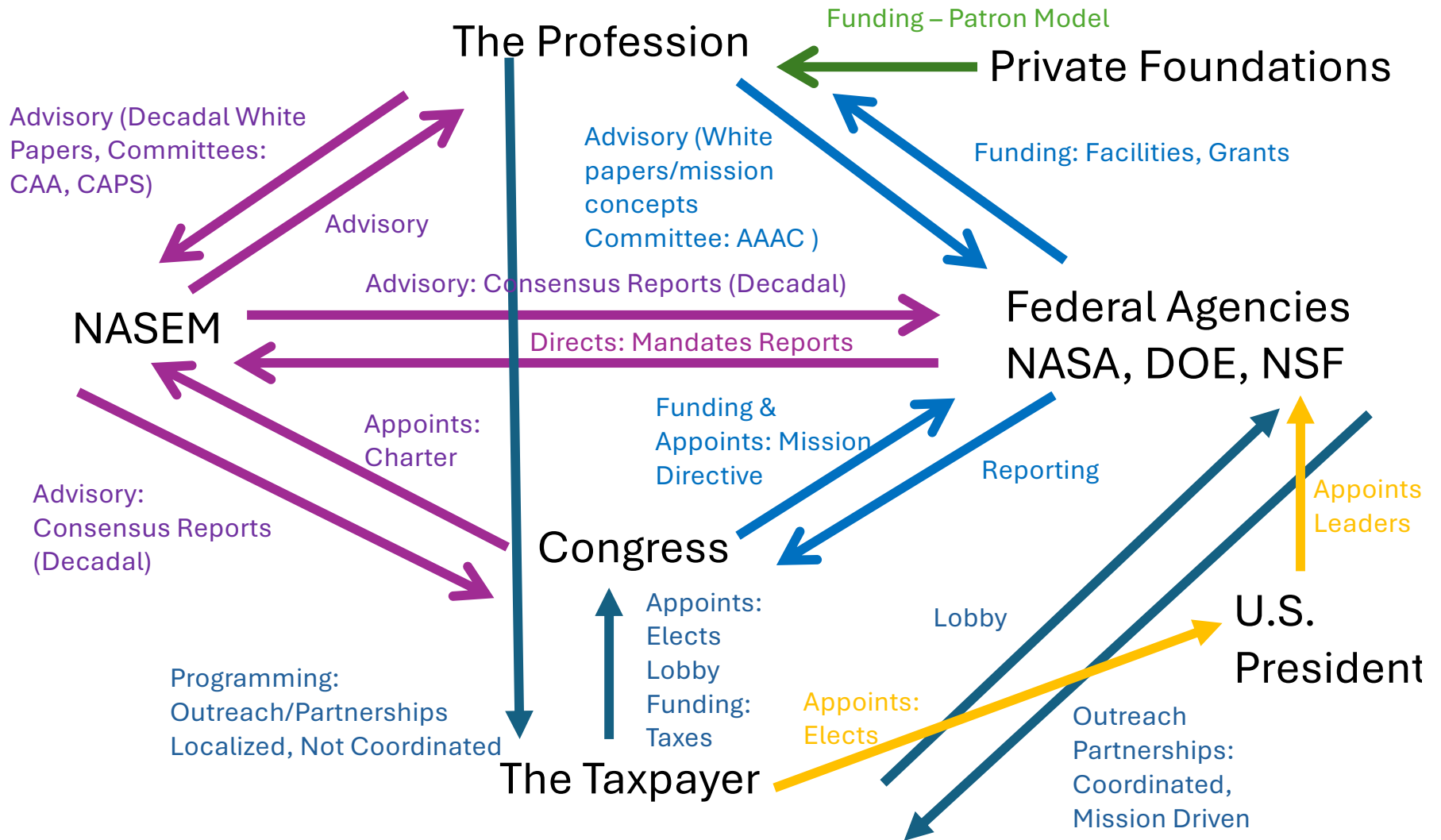
The Taxpayer











**The Flow of Decision Making from
The Profession to Congress**

Advisory (Decadal White
Papers, Committees:
CAA, CAPS)

NASEM

Advisory:
Consensus Reports
(Decadal)

Programming:
Outreach/Partnerships
Localized, Not Coordinated

The Profession

Advisory (White
papers/mission
concepts
Committee: AAAC)

Congress

Appoints:
Elects
Lobby
Funding:
Taxes

The Taxpayer

Funding – Patron Model

Private Foundations

Funding: Facilities, Grants

Federal Agencies
NASA, DOE, NSF

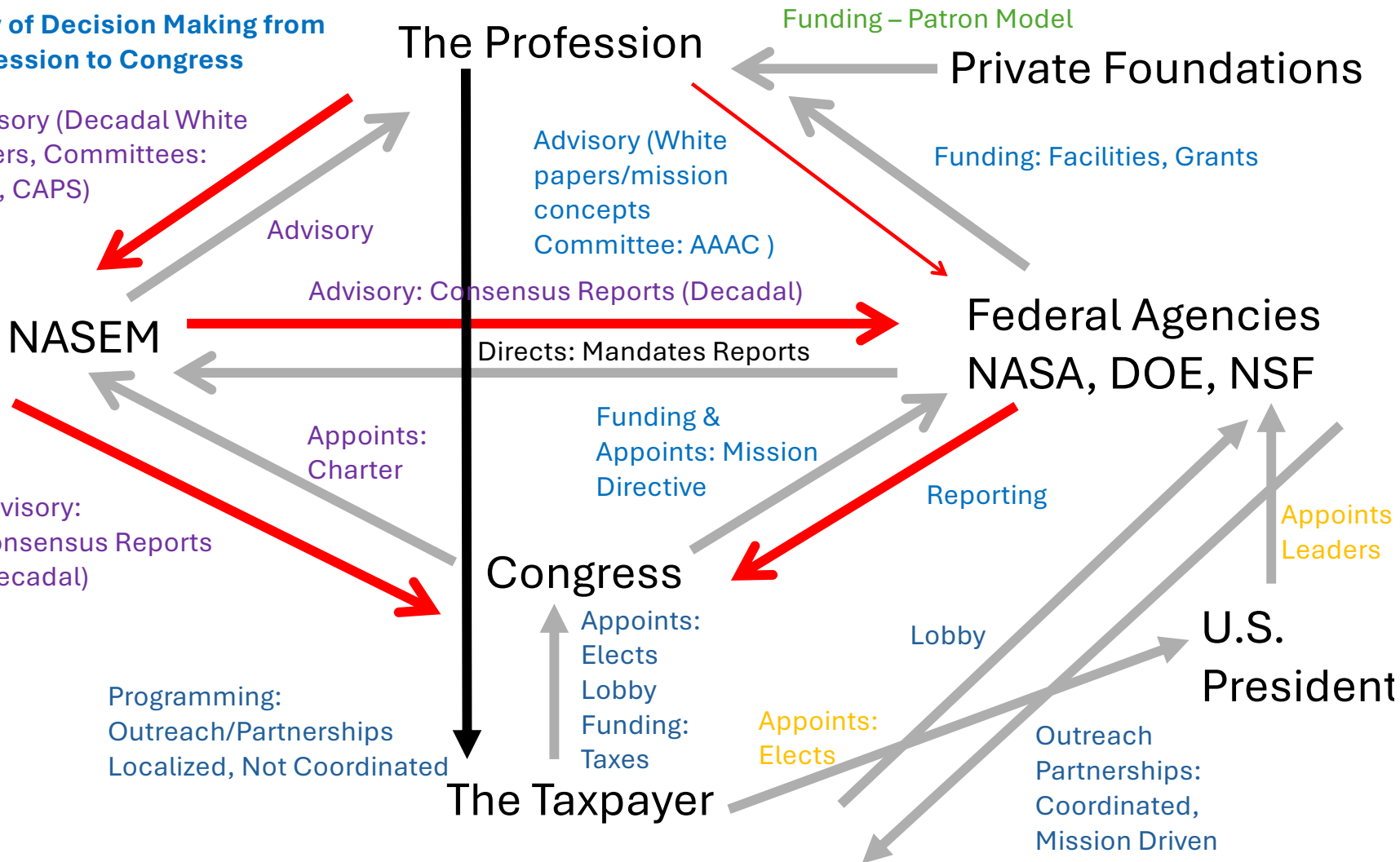
Reporting

Appoints
Leaders

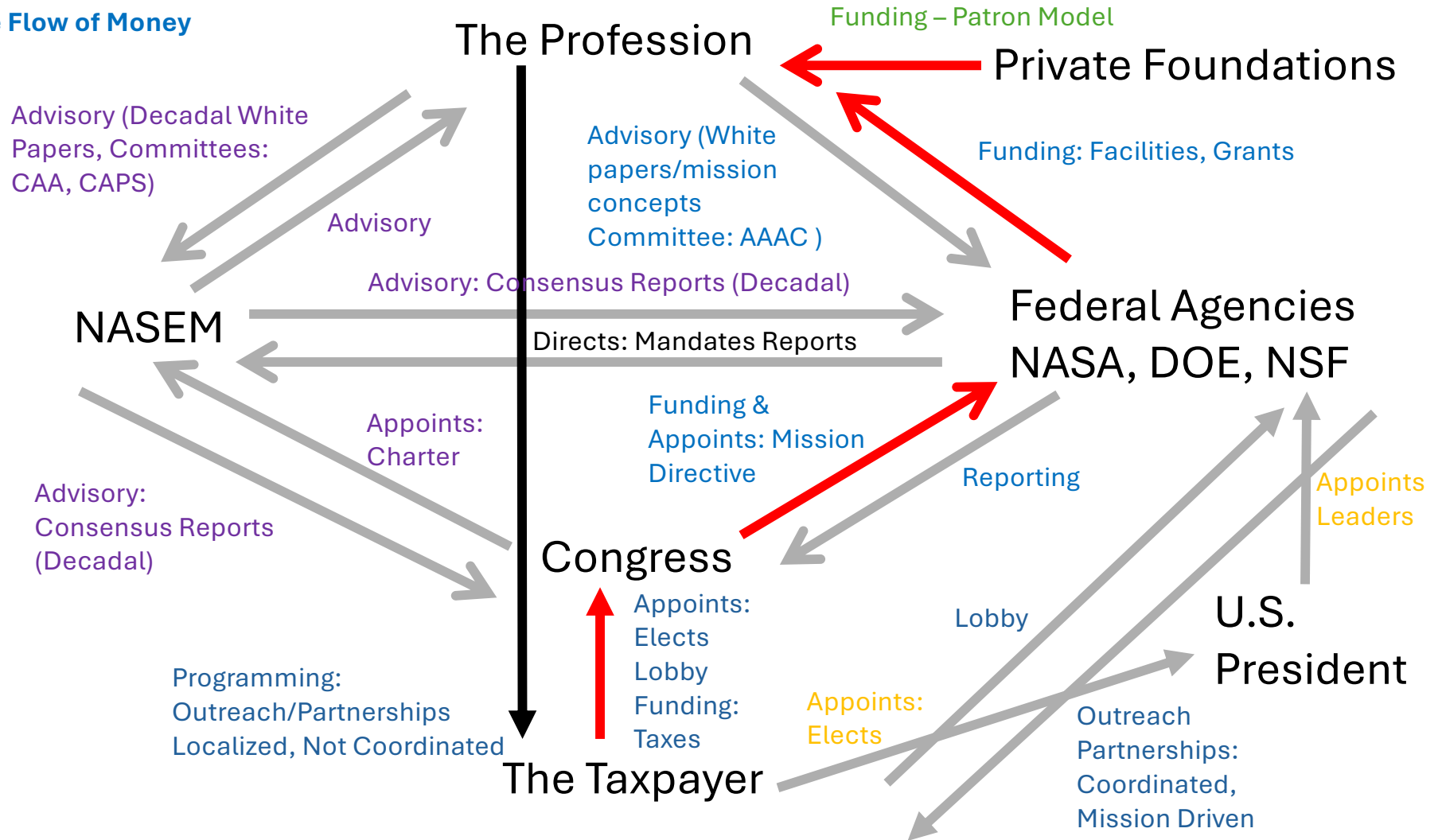
Lobby

U.S.
President

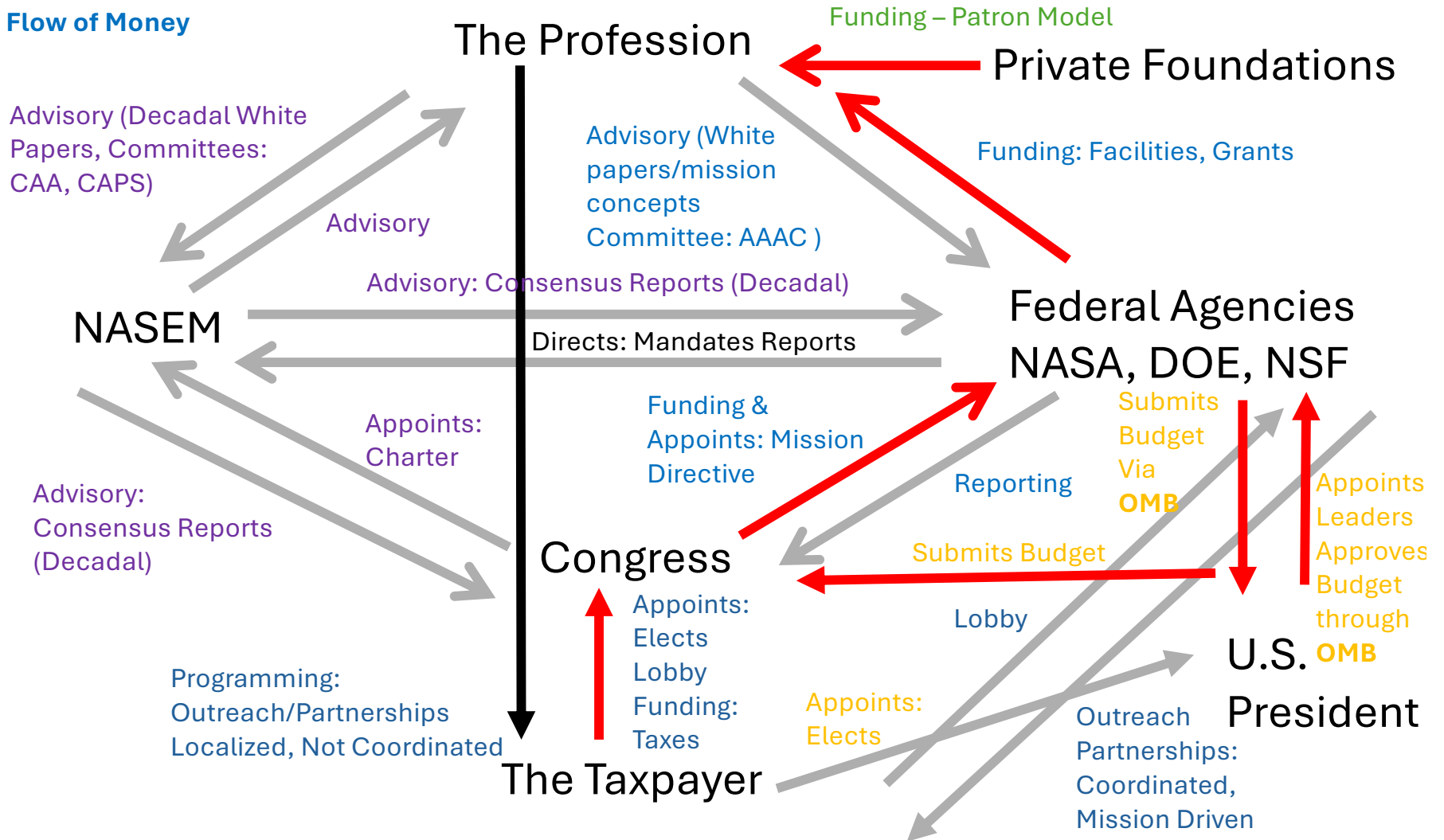
Outreach
Partnerships:
Coordinated,
Mission Driven



The Flow of Money



The Flow of Money



reference

- From Nick Schragal :
- Here is a link to the Astronomy on TAP LA presentation by Jack Kiraly of the Planetary Society discussing the 2026 budgeting process for primarily NASA but NSF & DOE in an astronomy context as well.
- <https://www.youtube.com/watch?v=NAOLEzSOV74>

Office of Management and Budget (OMB)

The Office of Management and Budget (OMB) serves the President of the United States in overseeing the implementation of his vision across the Executive Branch.

OMB shapes the budget before it is submitted to Congress as part of the President's Budget.

NASA/NSF/DOE Budgets are submitted to the OMB for approval by the President

Costs of Grad Students/Postdocs at UA

Federal Proposals (no IDC waiver)

55.5% (IDC as of 7/1/2026)

Graduate Research Assistant	FY27	FY28
Academic salary	\$25,654	\$26,424
Summer salary	\$16,700	\$17,201
ERE (Benefits: Health Insurance)	\$6,607	\$6,805
Tuition	\$15,500	\$16,740
IDC What is this?	\$27,173	\$27,989
Total	\$91,635	\$95,159

GRA Remarks: FY28 includes 3% salary increase and 8% tuition increase.

Postdoc*	Years of experience: (current NIH rates used for minimum postdoc salary levels)							
	0	1	2	3	4	5	6	7
Salary	\$62,232	\$62,652	\$63,120	\$65,640	\$67,824	\$70,344	\$72,960	\$75,564
ERE	\$9,708	\$9,774	\$9,847	\$10,240	\$10,581	\$10,974	\$11,382	\$11,788
IDC	\$14,388	\$14,485	\$14,593	\$15,176	\$15,681	\$16,264	\$16,868	\$17,470
Total	\$86,328	\$86,911	\$87,560	\$91,056	\$94,085	\$97,581	\$101,210	\$104,822

**Plus: relocation allowance and/or research budget*

Overhead/Indirect Costs (IDC)/Facilities & Administrative Costs (F&A)/ Indirect Cost Recovery (ICR)

- Expenses necessary to conduct the funded research, but are not linked to the specific research project (i.e. shared costs)
- E.g.
 - Facilities – building, office space, libraries
 - Local Computing Centers
 - Lab Equipment/Space
 - Compliance Support (Export Control)
 - IT Services
 - Utilities
 - Administrative Support/Business Offices

Overhead/Indirect Costs (IDC)/Facilities & Administrative Costs (F&A)/ Indirect Cost Recovery (ICR)

Origin:

- ICR Policy was developed during World War II - before this federal research funding was negligible.
- Agreement between the Federal Govt and Research Institutions that the government would offset costs of research that are not explicitly funded by the grant.
- **"No Profit, No Loss" Model:** incentivize research participation from universities, ensuring they broke even on critical projects like radar and the atomic bomb.
- Now, reasoning is *also* linked to patent development/Industry partnerships. But National Security is still a major driver [: e.g. UA](#)

Azoulay et al. March 2025 NBER (NIH ICR Policy Impact Assessment)

<https://www.nber.org/papers/w33627>

Pierre Azoulay is the *International Programs Professor of Management* at the MIT Sloan School of Management. His research focuses on the impact of different funding regimes on the rate and direction of scientific progress

Overhead/Indirect Costs (IDC)/Facilities & Administrative Costs (F&A)/ Indirect Cost Recovery (ICR)

- ICR Policy was developed during World War II (recall the link between NASEM & war effort).
- Agreement between the Federal Govt and Research Institutions that the government would offset costs of research that are not explicitly funded by the grant.
- “Government pays a set percentage point increment over the direct costs of research”
- This rate is *negotiated per institution*
- In 1966, the government removed indirect cost ceilings, enabling full **reimbursement** (ICR)

Overhead/Indirect Costs (IDC)/Facilities & Administrative Costs (F&A)/ Indirect Cost Recovery (ICR)

- WHY do we have this arrangement?

“The U.S. lacks a centrally-funded national university system (Dupree 1986). The second is a mismatch between the cost structure of science and the way research is supported by the U.S. federal government, which primarily provides project funding rather than institutional or infrastructure funding (Institute of Medicine 1990).”

Overhead/Indirect Costs (IDC)/Facilities & Administrative Costs (F&A)

- Why is ICR being scrutinized now? E.g. Proposed 15% Cap.
- ICR now appears to serve many purposes: as a (prospective) **incentive mechanism**, a (retrospective) reimbursement for overhead expenses, a way to encourage efficiency via cost-sharing, a means of providing financial slack, and more. This patchwork today is a contrast to ICR policy in the early postwar period, when research funding agencies each had specific goals their policies were designed around. The current range of goals or even disagreement over what the goals are may be an important driver of disagreement over the desirability of specific reforms and what their incidence and impacts might be.
- “Our analyses highlight the need for **greater transparency in indirect cost pools**, as the opacity of these expenditures limits our ability to assess their social value. Indirect cost proposals to HHS (the input to rate negotiations) are not publicly available.”

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The Flow of Money

- We are at a crossroads.
- Civil Servant Model or a Patron Model ?
- Can we find a balance between them?

Patron Model/Private Funding

- **Advantages:**
- **Disadvantages:**

Patron Model/Private Funding

- **Advantages:**

- Can allow researchers the freedom to pursue things that haven't been prioritized by the govt
- High Risk Science can be funded
- More independence (??????)
- Speed
- Stability (?)
- No IDC at all ???

- **Disadvantages:**

- If risky science also needs to be done – moves onus from fed govt
- Harder to find → find **someone** who cares what you are doing and you have to have **access to that person.**
- Potential conflicts – findings that contradict the goals of the funder → need to research the funder's values
 - Accountability
- Individualization without large scale vision of resources
- Stability (?)
- No IDC at all ?
- Pedigree matters for access.

Patron Model/Private Funding

- **Advantages:**

- Funding for high risk science
- Often less work to apply
- Can be a lot of money
- No IDC
- Once you are in the club, opens up more funding lines

- **Disadvantages:**

- Emphasis on Pedigree, leaves smaller institutions out
 - Who gets to be in the club? Access is not equitable
- Can be strings (Religious organizations, Where does the money come from?) - What are our values?
- Who is evaluating the proposals? And how ? Often no feedback, criteria unclear?
- Likely need lobbying/sensationalization.
- More uncertain – makes long term planning hard
- No coordination of efforts - no control or accountability structures